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A Literature Review on ERP Implementation: Methodologies, Module, Software, and Policy

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Abstract. This study examines the literature on Enterprise Resource Planning (ERP) system implementation methodologies, modules, software, and policies in recent years. ERP system investment has become the goal of many companies to improve company performance. An ERP system can be likened to the backbone of an information system in a company. The ERP system is integrated and can provide real-time information to aid decision-making. Various methodologies can be used in its implementation. Numerous ERP software options are also available, both paid and open source. Nineteen papers were collected from previous research on ERP system implementation and linked to discussion points. The study's findings indicate a uniqueness in ERP system implementation, which may differ from one implementation to the next. Furthermore, several open-source ERP systems are being considered, each with its own set of advantages and disadvantages. Open-source ERP systems are appropriate for small to medium-sized businesses. The use of modules is one of the factors to consider when choosing and implementing an ERP system.

Keywords: ERP; ERP Open-source, ERP Implementation

INTRODUCTION

In the last few decades, companies have invested in various aspects focusing on cost reduction, maximizing profits, and accuracy in decision making [1]. One of the investments made by many companies is the implementation of an Enterprise Resource Planning (ERP) system [2]. An ERP system is an integrated computer-based system designed to process part or all of a company's transactions and facilitate integrated planning in real time [3]. ERP can be analogous to the backbone of the information system in a company [4, 5]. Implementing an ERP system can result in higher quality, reduced time to market, improved communication, support in decision-making, shorter lead times, higher productivity, and lower costs [1]. ERP systems have been widely used in large-scale companies but are still limited to Small and Medium Enterprises (SMEs). Recently, many open-source ERP systems have been quite powerful and can be used by SMEs. Open-source ERP generally has a free license but limited modules and customization. [6].

Every ERP system implementation may have unique and different implementation methods [7]. The uniqueness of the implementation method depends on many things, such as the software used and the company's scale. According to Hietala and Päivärinta [8], although the ERP system can significantly improve organizational performance and business strategy, the challenges in its implementation are enormous. Therefore, ERP system implementation risks failure and even small companies are no exception. According to Svensson and Thoss [9], most of the causes of ERP system implementation failures are limited resources and a lack of human resource expertise in information technology. Many ERP systems can be used but have various advantages and disadvantages. Thus, choosing the right ERP system is important to ensure successful implementation and minimize risk.

ERP system users will be offered a package of modules that can be used; not infrequently, the modules are customized according to company needs. Many companies estimate the cost of an ERP project simply as a software

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license fee. Many issues must be considered in the ERP system implementation budget, such as software license, hardware, implementation, maintenance service, and training costs. ERP system implementation is an expensive process that increases with the company's size.

This paper presents a review of the implementation of the ERP system, which is divided into several aspects, namely, the type of ERP used is paid or free. Next is the ERP system implementation process in the company, the name of the ERP used, the size of the company that implemented the ERP, and lastly. The module used by the company. The remainder of this paper is organized as follows: Section 2 is for presenting the literature review. Section 3 is for ERP Software Category, ERP Implementation Methodologies in Section 4 and Section 5 for the conclusion.

LITERATURE REVIEW

Paper Classification of literature review can be seen in **TABLE 1**. Maguire, et al. [10] researched the implementation of the Oracle ERP system in telecommunications companies which are large-scale companies. Oracle is one of the paid ERP systems whose credibility is unquestionable. The implementation method begins with a project definition and then designs the basic procedures and infrastructure needed. After that, the configuration is carried out according to the company's business processes. Finally, test cases and post-implementation reviews are carried out before data migration.

Several previous studies on ERP system implementation have methodologies with the initial step of choosing an ERP system done by Sagala, et al. [11], Handoyono [12], and Damayanti and Cynthia [13]. After the ERP selection stage, integration and installation are carried out. Furthermore, an evaluation of the implemented system is carried out and continued with user training. The SAP ERP system used by the three researchers is included in the closed source or paid system category. Sagala, et al. [11] used various modules, including Sales, Production, Purchasing, Inventory, Warehouse, Invoicing, Distribution, Accounting, and Manufacturing, to implement in large-scale companies. Meanwhile, Handoyono [12] and Damayanti and Cynthia [13] implement ERP in medium-scale companies. Al-Mashari and Al-Mudimigh [14] researched the implementation of ERP SAP R/3 at a company called The Comp Group. The company is classified as large, and the ERP used is paid. The application method of this research begins with filling and alignment. Then a detailed conceptual design is carried out, followed by installation. In the end, an evaluation was carried out to assess the implemented system, ERP.

The SAP ERP system was chosen to be implemented in the research conducted by Falgenti and Pahlevi [15], Rahman [16], and Tarigan [17]. Falgenti and Pahlevi [15] researched the implementation of ERP SAP B1 in a medium-sized company. The application method of this research begins with the preparation stage and then the preparation of blueprints for further installation. In the end, testing of the implemented system and evaluation is carried out. In a different study by Rahman [16], SAP ERP system is implemented in large-scale companies. The company is engaged in broadcasting national television shows. The application method of the research begins with preparation and project determination activities. After that, blueprints are made. Realization and final preparation are the next steps before installation is carried out. The modules used in this case include Accounting, Material Management, Plant Maintenance, Human Resources, and Controlling. The implementation of the SAP ERP system in large-scale companies is also carried out by Tarigan [17]. In this study, a furniture, consumer good, and electronics company became the object of a case study. The ERP implementation method begins with identification, and then data collection is carried out. After that, the configuration and installation are executed and then evaluated.

According to 19 research articles, 58% of small, medium, and large-scale businesses used an open-source ERP system. This finding is exciting to be discussed further. It is known that all use of open-source ERP systems is aimed at small to medium-sized companies. In addition, the open-source ERP systems used include Odoo, Dolibarr, Ademppiere, Open Bravo, Front Accounting, and WebERP. Odoo's open-source ERP system is widely used in various ERP system implementation studies.

Hardjono, et al. [18], In their research, he implemented Odoo ERP with successive implementation methods (1) prepare infrastructure; (2) installation; (3) data preparation; (4) experiments; (5) testing; (6) evaluation. The modules implemented are Manufacturing, Controlling, and CRM. The application is carried out on small-scale companies engaged in the furniture sector. With the same implementation methodology, Ridho, et al. [19] Ridho, et al. [18] mengunakan Odoo ERP pada perusahaan berskala sedang. Module yang diterapkan adalah Sales, Purchasing, Warehouse, dan Manufaktur. Still using the same method, Fauzi, et al. [20], Fitrah, et al. [21], Aziza and Rahayu [22], and Akbar and Juliastrioza [23] implemented the Odoo ERP system in small-scale companies. The business fields are culinary business, electronics company, software engineering distribution services, and wholesale-retail. The Odoo ERP modules include Sales, Purchasing, Warehouse, Manufacturing, and Inventory. These various studies show that open-source ERP systems such as Odoo can cover common modules, which means they are sufficient for standard business processes.

TABLE 1. Paper Classification

Authors (Year)	Policy		Implementation Methodologies	ERP Software	Company Size			Module													
	Open Source System	Closed-Source			Small	Medium	Large	Sales	Production	Purchasing	Inventory	Warehouse	Invoicing	Distribution	Accounting	Manufacturing	Material Management	Plant Maintenance	Human Resource	Controlling	CRM
Maguire, et al. [10]		√	(1) project definition; (2) basic procedures; (3) infrastructure; (4) configuration; (5) test cases; (6) post-implementation review; (7) data migration; (8) go live	Oracle			√									√				√	√
Al-Mashari and Al-Mudimigh [14]		√	(1) filling and alignment; (2) conceptual detailed design; (3) installation; (4) evaluation	SAP			√	√	√	√	√										
Hardjono, et al. [18]	√		(1) prepare infrastructure; (2) installation; (3) data preparation; (4) experiment; (5) testing; (6) evaluation	Odoo	√			√				√									
Falgenti and Pahlevi [15]		√	(1) preparation; (2) blueprints; (3) installation; (4) testing; (5) evaluation	SAP		√		√		√	√				√						
Rahman [16]		√	(1) preparation (2) project; (3) blueprints; (4) realization; (5) final preparation; (6) installation and support	SAP			√								√		√	√	√	√	
Fauzi, et al. [20]	√		(1) prepare infrastructure; (2) installation; (3) data preparation; (4) experiment; (5) testing; (6) evaluation	Odoo	√											√					
Aziza and Rahayu [22]	√		(1) prepare infrastructure; (2) installation; (3) data preparation; (4) experiment; (5) testing; (6) evaluation	Odoo	√			√			√										
Putra and Azhari [23	√		(1) identification; (2) data collection; (3) analysis; (4) Installation, (5) data testing, (6) evaluation	ADempiere	√			√	√	√	√										
Tarigan [17]		√	(1) identification; (2) data collection; (3) configuration; (4) installation; (5) evaluation	SAP			√	√	√												
Akbar and Juliastrioza [23]	√		(1) prepare infrastructure; (2) installation; (3) data preparation; (4) experiment; (5) testing; (6) evaluation	Odoo	√			√		√	√										
Akbar and Akbar [24]	√		(1) identification, (2) installation; (3) testing; (4) evaluation	Openbravo		√		√	√		√										
Nofriandi and Kamil [25]	√		(1) identification, (2) installation; (3) testing; (4) evaluation	Front Accounting	√			√		√	√										

Still with an Open-source ERP system, apart from Odoo, Putra and Azhari [26] implemented the Dolibar Open-Source ERP system at a company engaged in processing clay, dolomite, and organic fertilizer. The company is classified as a medium-sized company, while the application method of the research begins with the identification stage and then makes a system proposal. After the two stages are executed, ERP application selection, configuration, and customization are carried out. In the final stage, implementation and testing are carried out. The modules implemented are Sales, Purchasing, Inventory, and CRM. Putra and Azhari [26] researched the implementation of ERP ADempiere in a furniture company. The company is classified as a small-scale company, and Adempiera is also one of the Open-Source ERP systems. This study's ERP system implementation method begins with the identification stage and then data collection. Furthermore, analysis is carried out and continued with the installation stage. In the final stage, data testing and evaluation are carried out. The modules applied are Sales, Production, Purchasing, and Invoicing.

Besides Odoo, Dolibar, and Adempiere, there are open-source ERP systems Openbravo, WebERP, and Front Accounting. Akbar and Akbar [24] implement Openbravo ERP in a medium-sized pharmaceutical company, and Nofriandi and Kamil [25] implement ERP Front Accounting in a small-scale minimarket. The implementation method used in both studies begins with the identification stage and then installation. After completing the installation stage, testing and evaluation are carried out before finally being able to go live. Another open-source ERP system that previous researchers have used is WebERP. AM and Akbar [27], researched the application of WebERP in a mobile shop. The company is classified as a small company. The application method of the research begins with the identification stage; then proceeds with preparation. After both processes are complete, then enter the installation stage. Finally, testing and evaluation are carried out before being able to go live.

A review revealed that several previous studies implemented open source ERP systems using various modules. Through in-depth analysis, it is known that most use is in manufacturing and retail with medium to low-scale companies. However, with various implementation methodologies, open source ERP is feasible to explore and implement in other business sectors such as service provider companies.

ERP SOFTWARE CATEGORY

Enterprise Resource Planning (ERP) is a system that can integrate all company functions information, including finance, sales, purchasing, human resources, and many more. According to Fernando, et al. [4] and Yang, et al. [5], ERP can be analogized as the backbone of an information system in a business. The ERP system is a development of Materials Requirement Planning (MRP) and Manufacturing Resource Planning (MRP II), which have existed since the 80s [28]. Pieces of information from different processes are put together in an ERP system. The integration offered in the ERP system can disseminate information in real time, thereby increasing accuracy for decision-makers [28]. In addition, ERP systems can improve business processes in an organization by reducing redundancy [29]. ERP systems have been widely used in large-scale companies but are still limited to Small and Medium Enterprises (SMEs). Recently, many open-source ERP systems have been quite powerful and can be used by SMEs. Open-source ERPs generally have a free license but limited modules and customization. [6].

According to Olson, et al. [30], Open Source System (OSS) has had good capabilities in the last decade. However, OSS can threaten various Close Source Systems (CSS). There have been more significant open source ERP systems developed in recent years. With more than 17,400 monthly downloads for one of the most popular Open Source ERPs, it can be said that this software is available and used by many companies. Lately, many large companies have been using open-source ERP systems. Therefore, it emphasizes that open-source ERP systems have a market. As the popularity of open-source ERP systems increases, so does the number of studies on open-source ERP. According to Wang and Wang [31], several open-source ERP systems recognized by the market include webERP, Compiere, PostBooks, Opentaps, OpenBravo, and OpenERP. In addition, currently, there is also an open-source Odoo ERP that quite exists. Open source ERP systems are suitable for small to medium-sized companies [32]. However, open-source ERP also has many disadvantages, with a low cost or no cost to purchase a system. In addition, there are also general obstacles to its implementation.

One of the disadvantages of open source software is limited software access. Companies that implement open source-based software systems cannot develop their systems. If the company wants to develop the software owned, the company must buy a reasonably expensive license. Therefore, on average, companies implementing ERP based on open source come from lower-middle companies. On the other hand, a paid large-scale company usually owns ERP. Therefore, although it has to incur a significant investment cost, it is directly proportional to the company's benefits.

ERP IMPLEMENTATION METHODOLOGIES

The successful implementation of an Enterprises Resource Planning (ERP) system will bring a competitive advantage. On the other hand, in extreme cases, implementation failure can cause the organization to go bankrupt [9]. Therefore, the implementation method is one of the critical factors in project success [33]. This section will describe the methodological stages of implementing several leading ERP systems, including SAP, Oracle, and Odoo ERP. The author will give a general description of implementing an ERP system for a company. Accelerated SAP (ASAP) is an implementation methodology suggested by SAP. According to Nagpal, et al. [33], ASAP method focuses more on straightforward implementation projects. The main stages in the ASAP ERP implementation method start with Stage 1: Project Preparation; this stage includes defining the project, identifying the scope and specifications, outlining the implementation strategy, and making a project schedule and implementation sequence to determine resources. Stage 2: Business Blueprint, at this stage, includes determining the requirements specification and data related to the master data. Stage 3: Configuration; this stage is carried out based on the business blueprint, and rigorous testing is required. Stage 4: Final Preparation; system testing and user testing are carried out at this stage, including user training. Stage 5: Go Live & Support, the stage where the user begins to use the implemented system.

Oracle Unified Method (OUM) is an implementation methodology recommended by Oracle. OUM consists of five stages based on the underlying principles advocated by leading industry standards. The main stages in the OUM ERP implementation method start with Stage 1: Inception; including precise synchronization of all stakeholders against project objectives. All of them are determined together with high-level requirements. Stage 2: Elaboration, detailed requirements development, and solutions. In this phase, prototype development can also be carried out for further analysis and development. Stage 3: Construction, a model is developed for system configuration according to operational standards plus customized components. The output of this stage is that the beta version of the system is released for further validation and acceptance testing. Stage 4: Transition, systematic testing, and validation are carried out so the organization can accept and use the new system. At this stage, the transition to the new system is also planned. Stage 5: Production, when the system comes into full use. At this stage, the system is monitored with full support. The Odoo Implementation Method is an implementation methodology recommended by the Odoo open-source ERP system. This method consists of five steps, starting with Stage 1: ROI Analysis, including doing ROI analysis, phasing & budget. Stage 2: Kick-Off, ROI analysis, phasing & budget. Stage 3: Implementation, Series of cycles: analysis, development, validation, key-user training. Stage 4: Go Live, End-user training, bug fixes. Stage 5: Second Deployment, broaden scope or add custom features.

CONCLUSIONS

Enterprise Resource Planning ERP is essential for companies to support their business processes. Currently, there are many choices of ERP systems. Closed Source Systems (CSS) and Open Source Systems (OSS) exist. Previous research found that open source ERP systems are generally used in small to medium-scale companies. Open-source ERP systems have a good market and are getting better and better. Of all the papers reviewed, more than 50% implement an open-source ERP system. There are various implementation methods proposed. SAP and Oracle's implementation method is a standard method that continues to be developed. For further research, look at the exciting open-source ERP systems. Researchers can focus on exploring reviews about open-source ERP systems.

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